

Lab Exercise 3

Lab Exercise: Selection Statements in C

Duration: 2 Hours

Objective

- To understand and implement conditional control structures in C.
- To apply logical decision-making in solving simple computational problems.
- To practice input handling using the Scanner class.

Task 1: Using if Statement

Exercise 1.1

Write a C program that accepts an integer number and checks if it is positive. Display "The number is positive" if true. Otherwise, do nothing.

Example Output:

```
Enter a number: 15
The number is positive
```

Exercise 1.2

Write a C program to read an integer and check whether it is even or odd. If it is even, print "The number is even." Otherwise, print ("The number is odd");

Example Output 1:

```
Enter a number: 8
The number is even
```

Example Output 2:

```
Enter a number: 17
The number is odd
```

Task 2: Using if...else Statement (with Multiple Conditions)

Exercise 2.1: Determine the Largest Number

Write a C program that reads three integers from the user and determines which number is the largest. Use `if...else` with multiple conditions (`&&`) to compare the numbers.

Example Output:

```
Enter first number    : 25
```

```
Enter second number : 30
Enter third number  : 18
The largest number is 30
```

Exercise 2.2: Check Student Eligibility

Write a C program to determine whether a student is eligible to take the final exam based on these conditions:

- The student must have an attendance of at least 80%
- The student must have a carry mark of 40 or more

Use if...else with logical AND (&&) and OR (||) operators to test both conditions.

Example Output 1:

```
Enter attendance percentage: 85
Enter carry mark: 45
Eligible to take the final exam
```

Example Output 2:

```
Enter attendance percentage: 75
Enter carry mark: 50
Not eligible to take the final exam
```

Task 3: Using interactive input and Conditional Statements

Write a C program that reads three test marks from the user, calculates the average, and determines the grade based on the following conditions:

Average Marks	Grade
80 - 100	A
60 - 79	B
40 - 59	C
Below 40	F

Requirements:

- Use if...else to determine the grade.
- Display both the average and the grade.

Example Output:

```
Enter mark for Test 1: 75
Enter mark for Test 2: 80
Enter mark for Test 3: 70
Average: 75.0
Grade: B
```

Task 4: Using Nested if Statement

Exercise 4.1: Student membership fee

A library charges subscription fees based on age and membership type:

If age ≥ 18 , then:

- If student, fee = RM10
- Otherwise, fee = RM20

If age < 18 , then:

- If has student card, fee = RM5
- Otherwise, fee = RM8

Sample Output 1 :

```
Enter your age :23
Enter your choice : Student (1) Not Student (2) : 1
Fee = RM10
```

Sample Output 1 :

```
Enter your age :15
Enter your choice : Student (1) Not Student (2) : 2
Fee = RM8
```

Sample Output 3 :

```
Enter your age :30
Enter your choice : Student (1) Not Student (2) : 2
Fee = RM20
```

Exercise 4.2: Loan Approval System

Write a C program that checks loan eligibility using nested if statements.

Conditions:

- The applicant must be employed.
- If employed, check the salary:
 - If salary $\geq 3000 \rightarrow$ Eligible for loan
 - otherwise $\rightarrow$ Not eligible
- If not employed $\rightarrow$ Not eligible

Sample Output 1:

```
Are you employed? (yes/no): yes
Enter your monthly salary: 3500
You are eligible for the loan.
```

Sample Output 2:

```
Are you employed? (yes/no): no
You are not eligible for the loan.
```